UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science – Software Engineering

ADVANCED WEB DEVELOPMENT

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Accounting System – Features Document

Product Backlog Decomposition into Features by Module

Features Summary

ID	Feature	Module	Sprints	Responsible	Total Hrs
FT-01	Access & User Management	Module 1	Sprint 1 (2 weeks)	E.Q / A.S / D.R	14 hrs
FT-02	Integration with SRI Online Portal	Module 2	Sprint 2 (1 week)	E.Q / A.S / D.R	14 hrs
FT-03	Invoice Download & Management from SRI	Module 3	Sprint 4 (1 week)	E.Q / A.S / D.R	14 hrs
FT-04	ATS Generation & Validation in XLSM	Module 4	Sprint 7 (1 week)	E.Q / A.S / D.R	19 hrs
FT-05	ATS Export to XML Format (DIMM SRI)	Module 6	Sprint 10 (2 weeks)	E.Q / A.S / D.R	14 hrs
FT-06	ATS Process Flow Traceability & Control	Module 7	Sprint 12 (1 week)	E.Q / A.S / D.R	14 hrs

Feature Details

FT-01 – Access & User Management

FT-01 – Access & User Management	
Description	Allows taxpayers to register with their RUC and log in securely to access all modules of the accounting system.
Objective	Ensure each user is uniquely identified by their RUC, controlling access in a secure manner.
Module	Module 1 – Registration and Authentication
Sprints	Sprint 1 (2 weeks)
Associated RFs	RF-01, RF-02

Responsible	Esteban Quiroga (T-01-03) | Angel Sabando (T-01-02, T-01-04) | David Rodriguez (T-01-01)
Business Rules	BR-02 (Taxpayer uniqueness)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-01	As a taxpayer I need to create an account linked to my RUC with my personal data.	Taxpayer	High	Prioritized
US-02	As a user I need to log in with my credentials to access my modules.	User	High	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-01-01	Create user registration form with respective validations.	US-01	David Rodriguez	Sprint 1	4
T-01-02	Model database to store users.	DBA	Angel Sabando	Sprint 1	4
T-01-03	Program login form with respective validations.	DBA	Esteban Quiroga	Sprint 1	3
T-01-04	Connect database with registration form.	DBA	Angel Sabando	Sprint 1	3

Applicable Non-Functional Requirements

NFR ID	Description
NFR-02	Intuitive interface with clear navigation between modules.
NFR-04	Encrypted HTTPS/TLS 1.2+ communication across all modules.
NFR-07	Compatible with Chrome, Firefox, and Edge (last 2 versions).

FT-02 – Integration with SRI Online Portal

FT-02 – Integration with SRI Online Portal	
Description	Connects the system to the 'SRI en linea' portal using taxpayer credentials, managing authentication, remote errors, and TLS security.
Objective	Establish a secure and reliable channel between the platform and the SRI to operate on behalf of the taxpayer.
Module	Module 2 – SRI Integration
Sprints	Sprint 2 (1 week) + Sprint 3 (1 week)
Associated RFs	RF-03
Responsible	Esteban Quiroga (T-02-01) | Angel Sabando (T-02-02) | David Rodriguez (T-02-03)

Business Rules	BR-02 (Uniqueness), BR-05 (SRI Updates)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-03 (F1)	As an accountant I need to link to SRI en linea with my credentials (basic authentication).	Accountant	High	Prioritized
US-03 (F2)	As an accountant I need the system to handle connection errors and apply TLS security.	Accountant	High	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-02-01	Create SRI connection interface using user credentials.	US-03	Esteban Quiroga	Sprint 2	7
T-02-02	Validate error handling and ensure login data is not leaked.	US-03	Angel Sabando	Sprint 3	5
T-02-03	Perform connection tests between the system and the SRI website.	US-03	David Rodriguez	Sprint 3	2

Applicable Non-Functional Requirements

NFR ID	Description
NFR-04	All communication with SRI must use HTTPS or TLS 1.2+.
NFR-05	The system must be available at least 99% of the time monthly.
NFR-08	The system must fail in less than 0.05% of executions.

FT-03 – Invoice Download & Management from SRI

FT-03 – Invoice Download & Management from SRI	
Description	Allows downloading invoices from the SRI filtered by tax period (monthly or semi-annual), storing them locally, and loading an XML directory for validation.
Objective	Automate invoice retrieval from SRI while respecting tax periods, eliminating manual downloads.
Module	Module 3 – Invoice Management
Sprints	Sprint 4 (1 week) + Sprint 5 (1 week) + Sprint 6 (2 weeks)
Associated RFs	RF-04, RF-05, RF-06
Responsible	Esteban Quiroga (T-03-01) | Angel Sabando (T-03-03) | David Rodriguez (T-03-02, T-03-04)
Business Rules	BR-01 (Tax periods), BR-02 (Uniqueness)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-04 (F1)	As an accountant I need to download invoices from the SRI (basic download).	Accountant	High	Prioritized
US-04 (F2)	As an accountant I need to filter invoices by month or semester (BR-01).	Accountant	High	Prioritized
US-05	As an accountant I need to load an XML invoice directory so the system can validate them.	Accountant	High	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-03-01	Create invoice download module linked to the user account.	US-04	Esteban Quiroga	Sprint 4	4
T-03-02	Develop a tax period filter system.	US-04	David Rodriguez	Sprint 5	4
T-03-03	Create a compatible directory for managing downloaded invoices.	US-04	Angel Sabando	Sprint 5	3
T-03-04	Create invoice directory upload and validation interface.	US-05	David Rodriguez	Sprint 6	3

Applicable Non-Functional Requirements

NFR ID	Description
NFR-01	Support processing of more than 5,000 invoices per month.
NFR-03	Generate the ATS in a maximum of 5 seconds with up to 1,000 invoices.
NFR-06	Format updates without affecting other modules.

FT-04 – ATS Generation & Validation in XLSM

FT-04 – ATS Generation & Validation in XLSM	
Description	Automatically generates the ATS in XLSM according to the SRI schema, validates each field, and produces an error log for correction before exporting to XML.
Objective	Reduce typing errors and ensure the ATS complies with the SRI technical specification before being exported.
Module	Module 4 – XLSM Generation / Module 5 – XLSM Validation
Sprints	Sprint 7 (1 week) + Sprint 8 (2 weeks) + Sprint 9 (2 weeks)
Associated RFs	RF-07, RF-08
Responsible	Esteban Quiroga (T-04-01, T-05-01) | Angel Sabando (T-04-02, T-05-02) | David Rodriguez (T-04-03)
Business Rules	BR-02 (Uniqueness), BR-03 (ATS format and validation)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-06 (F1)	As an accountant I need to generate the ATS in XLSM from the loaded invoices.	Accountant	High	Prioritized
US-07	As an accountant I need to edit and validate the XLSM to detect errors before the final XML.	Accountant	High	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-04-01	Transfer invoices to ATS XLSM format validating data transcription.	US-06	Esteban Quiroga	Sprint 7	5
T-04-02	Validate each ATS field according to invoice XML tags.	US-06	Angel Sabando	Sprint 8	5
T-04-03	Enable download of the generated ATS file in XLSM.	US-06	David Rodriguez	Sprint 8	2
T-05-01	Develop full validation of the XLSM file.	US-07	Esteban Quiroga	Sprint 9	5
T-05-02	Design error log system and enable XML creation.	US-07	Angel Sabando	Sprint 9	2

Applicable Non-Functional Requirements

NFR ID	Description
NFR-03	ATS generation in a maximum of 5 seconds (up to 1,000 invoices).
NFR-06	ATS format updates without affecting other modules.
NFR-08	Failure rate below 0.05% in ATS generation.

FT-05 – ATS Export to XML Format (DIMM SRI)

FT-05 – ATS Export to XML Format (DIMM SRI)	
Description	Converts the validated XLSM file to the XML format required by the SRI DIMM in UTF-8 encoding and makes it available for download and official submission.
Objective	Generate the final XML file complying with the SRI DIMM schema, ready for official submission.
Module	Module 6 – XML Export
Sprints	Sprint 10 (2 weeks) + Sprint 11 (2 weeks)
Associated RFs	RF-09
Responsible	Esteban Quiroga (T-06-01) | Angel Sabando (T-06-02) | David Rodriguez (T-06-03)
Business Rules	BR-04 (XML format required by SRI), BR-05 (SRI Updates)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-08 (F1)	As an accountant I need to generate the ATS in XML UTF-8 from the validated XLSM.	Accountant	High	Prioritized
US-08 (F2)	As an accountant I need the XML to be compatible with the SRI DIMM schema.	Accountant	High	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-06-01	Develop XML creation from the validated XLSM.	US-08	Esteban Quiroga	Sprint 10	5
T-06-02	Validate that the XML complies with the SRI DIMM schema.	US-08	Angel Sabando	Sprint 11	5
T-06-03	Implement XML download with applied validations.	US-08	David Rodriguez	Sprint 11	4

Applicable Non-Functional Requirements

NFR ID	Description
NFR-01	Support large volume exports without degrading performance.
NFR-04	Encrypted HTTPS/TLS communication during the export process.
NFR-08	Failure rate below 0.05% in XML generation.

FT-06 – ATS Process Flow Traceability & Control

FT-06 – ATS Process Flow Traceability & Control	
Description	Provides a control panel showing the status of each step in the ATS process: from SRI linking to XML export, with date and action records.
Objective	Ensure full visibility and audit of the workflow, facilitating detection of incomplete steps.
Module	Module 7 – Flow Control / Traceability
Sprints	Sprint 12 (1 week)
Associated RFs	RF-05 (ATS Manager Module)
Responsible	Esteban Quiroga (T-07-01) | Angel Sabando (T-07-02)
Business Rules	BR-03 (Format and validation), BR-05 (SRI Updates)

Associated User Stories

ID	User Story	Actor	Priority	Status
US-09	As an auditor I need to view the complete history of the ATS generation process.	Auditor/Admin	Medium	Prioritized

Technical Tasks

Task ID	Task	Rel. with	Responsible	Sprint	Hrs
T-07-01	Create widget with ATS process steps and their status (completed / pending / blocked).	US-09	Esteban Quiroga	Sprint 12	7
T-07-02	Connect widget to each module for automatic update upon task completion.	US-09	Angel Sabando	Sprint 12	7

Applicable Non-Functional Requirements

NFR ID	Description
NFR-02	Intuitive interface with clear navigation and minimal task execution times.
NFR-05	99% monthly availability to ensure continuous access to the history.
NFR-07	Compatible with Chrome, Firefox, and Edge (last 2 versions).

Effort Distribution by Team Member

Team Member	Assigned Tasks	Estimated Total
Esteban Quiroga	T-01-03, T-02-01, T-03-01, T-04-01, T-05-01, T-06-01, T-07-01	36 hrs
Angel Sabando	T-01-02, T-01-04, T-02-02, T-03-03, T-04-02, T-05-02, T-06-02, T-07-02	34 hrs
David Rodriguez	T-01-01, T-02-03, T-03-02, T-03-04, T-04-03, T-06-03	30 hrs
PROJECT TOTAL	**21 tasks distributed across 12 sprints**	**100 hrs**